

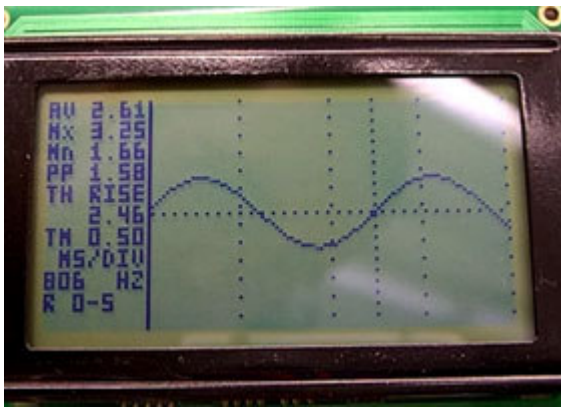
SEMIFLUID

sem-i-flu-id - adjective - Intermediate in flow properties between solids and liquids;

PIC18F2550 KS0108 Graphical LCD Oscilloscope

August 21, 2006

Filed under: [PIC Projects](#), [C Projects](#), [PIC18F2550](#), [GLCD](#) — Steve



Using the [PIC18F2550 GLCD Text Test](#) as a basis for further together a simple and accurate graphical oscilloscope using a microcontroller and a [AGM1264F](#) graphical LCD. The oscilloscope displays the [average](#) voltage, the [maximum](#) voltage, the [minimum](#) voltage, and the [zero-crossing](#) frequency for a DC signal over time. The oscilloscope has a built in [edge trigger](#) function that can be set to trigger on a rising (or disabled altogether). The time scale for the display is easily redefined using the changeTimeDivision function. Likewise, the voltage range can be changed to 0-5V, 0-2.5V, and 0-1.25V. The main features of the oscilloscope include relatively slow acquisition time and [sensitivity](#) and the fact that the inputs are limited by the constraints of the microcontroller. However, it is a very nice display and I highly suggest you view

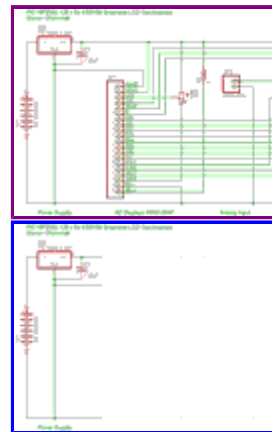
in action.

Full Circuit

The source and firmware for the circuit can be found at the bottom of the page. Each section of the circuit is labeled in the schematic. All of the sections and their components are described and discussed below. The part numbers for the components are linked to websites for data and more information when available.

Power Supply

The power supply uses a 9 volt battery and a [TC1262-5.0V](#) high-accuracy low-dropout linear voltage regulator to provide a stable 5 volt supply for the microcontroller and the graphical LCD. A 1uF (microFarad) polarized decoupling capacitor is necessary on the output of the voltage regulator to prevent power spikes or ripples. A [wall wart](#) power supply as low as 5.3V can be substituted for the 9 Volt battery.



AZ Displays AGM1264F

The [AGM1264F](#) is 128 x 64 pixel graphical LCD with built in controllers that allow it to be easily controlled using a PIC microcontroller. It is a 128 x 64 pixel graphical LCD and has an LED backlight, onboard negative voltage generator (7660) for LCD driving, onboard Samsung [KS0108](#) controller, and an 8-Bit MPU interface.



Analog Input

The 1st analog input (Pin A0) is configured to be an analog input. Please note that “The source impedance affects the offset voltage at the analog input (due to pin leakage current). The maximum recommended impedance for analog sources is 2.5 k Ω .” ([PIC18F2550](#), p.260)



Microcontroller

The microcontroller used is a Microchip [PIC18F2550](#). I modified the PIC18F2550 [Tiny PIC Bootloader](#) assembly file so I could use a 20MHz crystal/resonator at 115,200 baud (the modified bootloader can be found at the bottom of the page). The PIC18F2550 runs at 48MHz using the internal [PLL](#). R1 is a pull-up resistor necessary for operation. C1 is a stabilizing capacitor that is used for the onboard USB voltage regulator, which is not used in this project. The component marked ‘RES’ is a 20MHz resonator.

RS232 Level Converter

The microcontroller USART pins need to be connected to a [RS-232 Level Converter](#) to connect to a PC for firmware updates using the Tiny PIC Bootloader. Otherwise, after initial programming they can be left disconnected.



Source and Firmware

The PIC must initially be programmed with the ‘SAC_tinybld18F2550usb_20MHz_115200_48MHz’ hex file to program the bootloader on the PIC. Then, using Tiny PIC Bootloader placed on the chip using the Tiny PIC Bootloader [frontend](#) with ‘12h 34h 56h 78h 90h’ in the ‘List of code’ menu. Please feel free to [contact me](#) if you have any problems.

- [SAC_tinybld18F2550usb_20MHz_115200_48MHz.HEX](#)
- [18F2550 GLCD Oscilloscope DR1r1.c \(hex\)](#)
- [GLCD - modified.c](#)

Videos

[Here](#) are three videos of the PIC18F2550 GLCD Oscilloscope in action. The first video demonstrates sine waves of different frequencies. The second video demonstrates different waves with shifting amplitudes. The third video compares the display of the PIC18F2550 GLCD Oscilloscope to the display of a [Hewlett Packard](#).

Update (November 15, 2006)

Here is the firmware without the bootloader code:

- [18F2550 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)

Update (February 1, 2007)

Here is untested firmware for the PIC18F4550 without the bootloader code:

- [18F4550 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)

Update (April 11, 2007)

Here is untested firmware for the PIC18F252 and PIC18F452 without the bootloader code. Please note that it uses a 10MHz crystal/resonator instead of a 20MHz one.

- [18F252 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)
- [18F452 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)

Bookmark to:



Comments »

1. Would you please advise the Full Part number of AGM1264F?

Comment by Ricky — November 26, 2006 @ [7:53 am](#)

2. The display I have is an AGM1264F-FL-GTS-D, which is (according to AZ Displays, <http://www.azdisplays.com/prod/g1264f.php>):
 - o Polarizer Type: Transflective
 - o Backlight Type: Yellow LED
 - o Fluid type: Grey mode STN
 - o Viewing Direction: Top view (12 o'clock)
 - o Temperature Range and Power Supply: Standard temperature range requiring a single power

I purchased it from BG Micro: <http://www.bgmicro.com/prodinfo.asp?prodid=LCD1030>

Comment by [Steve](#) — November 27, 2006 @ [1:37 pm](#)

3. Hi! Can I use PIC18f4550 instead of PIC18F2550 in scope project?
Guntis

Comment by Guntis — December 5, 2006 @ [1:13 am](#)

4. Guntis - Yes, you can use a PIC18F4550. The architecture is the same, as is the memory mapping. is double check the pin definitions and change the device include at the top of the source code to: `<pic18f4550.h>` before you recompile.

Comment by [Steve](#) — December 6, 2006 @ [12:08 pm](#)

5. It's don't work on my GLCD: type: tech12864g
see the pins of LCD:

1-vss
2-vdd
3-v0
4-rs
5-r/w
6-e
7-b0
8-b1
9-b2
10-b3
11-b4
12-b5

13-b6
 14-b7
 15-cs1
 16-rst
 17-vee
 18-cs2
 19-backlight
 20-backlight

Comment by [Luis Fernando](#) — December 6, 2006 @ [8:10 pm](#)

6. Luis - I checked a datasheet for the [TECH12864G](#) and noticed that the display may not have a neg for VEE and VO. The LCD should work with this schematic (other than the different pinout) beca same. Can you check to see if the output of VEE is about -10V?

Comment by [Steve](#) — December 11, 2006 @ [11:36 am](#)

7. hi

please i want code in picbasic language

Comment by [alaaodeh](#) — December 18, 2006 @ [9:29 am](#)

8. alaaodeh - Sorry, I've never coded in picbasic. Please feel free to port my code though!

Comment by [Steve](#) — December 18, 2006 @ [11:48 am](#)

9. Nice work ! Did you plan to use an external A/D converter ? a serial one, maybe... How about Lo

Comment by [Nicu](#) — December 19, 2006 @ [3:14 pm](#)

10. Hi,
 with the firmware without bootloader (Update 15 Nov 2006), I program the pic and it works? Notl project!!!

Comment by [Mimmo](#) — December 20, 2006 @ [4:32 am](#)

11. Nicu - An external A/D converter would help increase the sampling rate. Right now the oscillosco oscilloscope. I could greatly increase the speed and resolution of the oscilloscope if I used an exte it as a logic analyzer, to be honest I hadn't considered it. A logic analyzer would be very easy to p The only issue that I could think of is whether or not the resolution of the LCD (128x64) is high e channels.

Mimmo - For the firmware without bootloader, yep, all you need to do is program the PIC, make t connections, and it will work! Please note that this version of the firmware has a set resolution and building an oscilloscope with a rotary encoder and pushbutton to change settings.

Comment by [Steve](#) — December 20, 2006 @ [2:41 pm](#)

12. Thanks Steve,
I cannot understand C, only assembler and PicBasiPro, so I am not able to modify your C firm to r
a rotary encoder. I will wait your next relase, hi. I have all materials, so I can start to build it.
It would be nice to use an external A/D converter to increase performaces from a simple “logic an
“oscilloscope” even if with only one channel. Congrats again.
Ciao from Rome - Italy

Comment by Mimmo — December 20, 2006 @ [4:41 pm](#)

13. How about using Microchip USB bootloader ? I think that 128×64 pixels should be ok for Logic A
one could be used 😊

Comment by Nicu — December 21, 2006 @ [11:43 am](#)

14. I also think it owuld be very nice to have the code available in BASIC or pseudo code. This way it
to other uC's. I am more of an AVR guy :p Maybe even just a flow chart of the process? If you ev
you let me know! I'd really be interested!

Comment by Matthieu — January 13, 2007 @ [3:59 pm](#)

15. Matthieu - I haven't ported the code to BASIC or pseudo code, but that is definitely a future possil
should be ANSI C (although it includes a number of functions that are internally defined), so it sh
it to an AVR (using [avr-gcc](#)). I'll keep updating and see what I can do!

Comment by [Steve](#) — January 14, 2007 @ [1:54 pm](#)

16. Steve, for work with slow indicators, such as XIAMEN GDH12864A and similar, in standard func
reading in LCD () is necessary to add file GLCD.C a delay on 2 uS between distribution of the dat
LCD/RC2) and also to increase on 2 uS duration of it crпoбa. Otherwise the screen “collapses”. 😊
Yours faithfully, Andrey. Moscow.

Comment by Andrey — January 15, 2007 @ [2:51 pm](#)

17. Andrey - Some displays are slower than others, so yes, it would be beneficial to use the delay. On
(AGM1264F) the screen updates perfectly fine without it; however, that does not mean that every
will. I have found that some HD44780 compatible drivers (text LCD drivers) have similar issues v
speed, so you need to insert delays to ensure that there is no display corruption.

Comment by [Steve](#) — January 16, 2007 @ [10:46 am](#)

18. Steve, Are you working on the new version of this little jewel? I've read you are planning to make
buttons and rotary encoder to change setting. Do you think I can start to build this one and then up
firmware and wiring modifications as it will be available?
Are there any issue on 18f2550 programming? Will a standard JDM programmer work with this p
Thanks
Francesco

Comment by Francesco — January 20, 2007 @ [10:37 am](#)

19. Francesco - Yes I am, but very slowly. I have been trying to complete human subject testing for or and it has taken much more time than either of us expected. It has been a long week (and weekend include:

1. Auto-ranging (both voltage and time domain)
2. 3 pushbutton for manually changing parameters
3. USB support, or at the very least USB bootloading

If you can program the firmware initially then there should not be a problem in the future. Also, as programmer supports the PIC18F series (and specifically the PIC18F2550), then there will not be online and found a couple of sites that say they support both the JDM programmer and the PIC18F (<http://www.members.aon.at/electronics/pic/picpgm/index.html>, <http://www.qsl.net/dl4yhf/winpic> seems like you might need to slightly modify the programming circuit: <http://www.sparkfun.com/bin/phpbb/viewtopic.php?p=3572#3572>

Comment by [Steve](#) — January 21, 2007 @ [8:55 pm](#)

20. Please, can someone ported the originally code to “MikroC” or “MikroBasic” from Mikroelektron
Thanks
Vincenzo

Comment by Vincenzo — January 24, 2007 @ [9:55 am](#)

21. Vincenzo - I have not programmed using MikroC or MikroBasic, so unfortunately I can not easily
However, if MikroC is an ANSI C compiler, then porting the program from CCS C should not be

Comment by [Steve](#) — January 25, 2007 @ [12:52 pm](#)

22. Hi Steve, Great job with this! I have a CREATE USB Interface with the PIC18F4550 and a LCD, compiler... could you post a '4550 hex file (no bootloader) for me?

Thanks
-Dan

Comment by Dan — January 31, 2007 @ [5:21 pm](#)

23. #22 (Dan) - I am more than happy to provide a copy of the [code](#) and compiled [hex](#) for the PIC18F have though is if the CREATE USB interface requires firmware compiled to include ORG inform: bootloader. If not, then hopefully this will work without a problem. Please let me know!

Comment by [Steve](#) — February 1, 2007 @ [10:23 am](#)

24. Hi Steve - Thanks! I'm using an ICD2 to program the CUI, so I'll just overwrite the bootloader fo back on)... Normally though, the USB bootloader resides in 0x0000,0x0800.

Thanks again for the compiled hex - it's too bad the CCS compiler demo doesn't work on the PIC have ot buy it some day...

Cheers,
-Dan

Comment by Dan — February 1, 2007 @ [2:51 pm](#)

25. Hi Steven,
Your project looks great!! I love interesting projects like this. I emailed you a message about possible project for a senior design project. I would like your permission to perform something of that capacity at your decision.

A prompt response will be greatly appreciated.

P.S. My email address is in your inbox I asked you a question from www.semifluid.com.

Thanks!

Comment by Will — February 8, 2007 @ [12:20 am](#)

26. Steve - Please check your mail. By the way I want to know the value of R3 used in circuit. And, for the bootloader, how can I erase the program code only without erase the bootloader code. Is the tiny bootloader program the program code automatically after press "Flash".

Comment by Ricky — February 9, 2007 @ [4:22 am](#)

27. #25 (Will) - Sorry for the delay in my response, I emailed and let you know that you are more than welcome to use my projects. All of my projects are licensed under the Creative Commons Attribution-Share Alike 2.5 (<http://creativecommons.org/licenses/by-sa/2.5/>), although I don't clearly state it on my webpage (that's in the footer...). All that I ask is that you include me in your credits and publish the project in your blog.

#26 (Ricky) - I'm catching up with emails right now. The value of R3 was intentionally left blank and dependent upon your backlight. You can either use a potentiometer or a fixed resistor to set the brightness of the backlight. Check the datasheet for your LCD and you can choose an appropriate resistor given the voltage drop of the LEDs. Tiny PIC Bootloader automatically erases the necessary program space so you do not need to worry about overwriting or erasing the bootloader because the PC frontend makes sure to do so for any incompatibility (and it will tell you if there is a problem with the memory space!).

Comment by [Steve](#) — February 9, 2007 @ [2:06 pm](#)

28. Steve - Long time no see, how are you?

Finally, I got the ICD2 and programmed the "SAC_tinybld18F2550usb_20MHz_115200_48MHz.HEX".

However, when I use Tiny bootloader from <http://www.etc.ugal.ro/cchiculita/software/picbootloader/> to program the program to my PIC.

I did the following procedures:

- 1) Connected the circuit as file per attached
- 2) Set PC COM1 Config: 115200, 8 bits, Hardware control: Hardware, use FIFO buffer

- 3) Executed Tiny bootloader
- 4) Connected COM1 with circuit per attached
- 5) Entered '12h34h56h78h90h' in the 'List of codes to send first:'
- 6) Press "writeflash", and press switch on circuit as quick as possible

After taht the following message displayed:

```
Connect COM1: ok
Wait 100ms
HEX: 35 days old, INHX32, 18F code+cfg, total=15548bytes
Searching for PIC ...Not found,
ERROR!
```

And, sometime tiny bootloader can identified my PIC is PIC18F2550, but it will display "can't wr

Would you please help to check is that any wrong in my circuit per attached, or any other solution to program on the IC directly?

The following link is the circuit:

<http://rickylab.tripod.com/tinubootloadercircuit001.jpg>

Comment by Ricky — February 20, 2007 @ [12:51 pm](#)

29. Hi everbody,
I prepared a PCB for this oscilloscope and I soldered the component on the PCB. When I powered generator), I saw the signal on the GLCD but it show only square wave signal. Then I realize that nearly 4,90volt and pk-pk is not changing. I adjust the waveform generator as a sine wave, pk-pk 1000HZ. I saw righth value as 1000Hz but it still show square wave and max 4,86v. Where is the p
Thanks...

Comment by Baris — February 22, 2007 @ [3:39 pm](#)

30. hi.this designs is very attractive.
i want to build this oscilloscope.but i am a amateur,so i have 3 questions.
1-which one from this files shode put in ic?
2-to change the range of voltage and time there is no key?
3-my glcd is TS12864A-2.this lcd work buy KS0108B and KS0107B processor.can i use thise lcd

Comment by reihaneh — March 1, 2007 @ [4:43 am](#)

31. Very interesting project!

I want to get one. Does anyone have this DIY project combined with PIC18F2550 USB HID Osci
Please email to me.

rafkumar@hotmail.com

Comment by Raf Kumar — March 4, 2007 @ [8:58 am](#)

32. HI, THANKS FOR THIS PROJECT, I HAVE A QUESTION IS POSSIBLY MAKE THIS READ VOLTAGE RMS?, THANKS...

Comment by [GIOVANNI](#) — March 8, 2007 @ [7:12 pm](#)

33. What does video look like? I would be interested in making a basic waveform monitor and this loc starting point

Comment by [Uncle Chris](#) — March 16, 2007 @ [5:05 pm](#)

34. I was wondering how you went about deriving your time delays in the sampling code? I want to p 16f877A (which is also supported by the boot loader) but instead of displaying to the lcd I would l The 877a allows up to 20mhz so I will probably need to adjust the delay values. Could you provid Thanks in advance!

Comment by [ScottM](#) — March 31, 2007 @ [1:21 am](#)

35. hi, this project is great.
i dont have p18f4550 but i have p18f452 please can you post a hex file (no bootloader) for pic18f4

Comment by [Ahm1](#) — April 4, 2007 @ [1:38 pm](#)

36. #32 (GIOVANNI) - This currently reads frequency using zero-crossing detection. It could potentia DFT or FFT algorithm.

#33 (Uncle Chris) - Honestly, I'm not sure, I should hook it up and see what the output looks like!

#34 (ScottM) - The time delays are functions in CCS C. I've verified (using a seperate oscilloscope measurements are very close to the actual signal, but given that I am using an imprecise timebase i can not presume that they are exact.

#35 (Ahm1) - Ask and you shall receive! See the update at the bottom of the post, it includes comp PIC18F252 and PIC18F452:

- o [18F252 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)
- o [18F452 GLCD Oscilloscope DR1r1 - no bootloader.c \(hex\)](#)

Comment by [Steve](#) — April 11, 2007 @ [4:41 pm](#)

37. Hi, I'm going to use the PIC18F4620, would that be ok? I will ofcourse include the correct header voltage regulator has a max input voltage of 6v (from the datasheet) did you have any problems i

Cheers...

Comment by [Ashley](#) — April 15, 2007 @ [10:57 am](#)

38. Hi,
I was wondering what it is the maximimun frequency that you are able to use with this oscilloscope

Comment by Micky — April 21, 2007 @ [7:51 am](#)

39. Great project Steve,

I have been trying to get my 64*128 to display a sine wave for ages and just don't have the know l difficulty trying to work out the process to plot the pixel points, too much math for me I think. I N work to 20kHz and your project looks spot on. Do you have a process or flow chart for how the si works please ?

Comment by Gordon — April 28, 2007 @ [5:32 pm](#)

40. #39 (Gordon) - I dropped you an email about this, but I should post it as well. Unfortunately I don't have a flow chart however here is a basic rundown of what the Graphical LCD O-scope does:

1. Samples all of the analog-to-digital converter (ADC) at a predefined rate and stores 100 samples.
2. The discrete samples are then multiplied by a constant ratio so that they will fit on the screen. The samples are converted from a range of 0-255 to 0-63.
3. The samples are then plotted one by one.

I'd be happy to help you out further if you have questions about it. The actual display of the sinusoid is a simple x-y plot. The sinusoidal wave is actually what a function generator is presenting to the ADC.

Comment by [Steve](#) — April 29, 2007 @ [2:50 pm](#)

41. help me with X input of this qscilloscope
best regards
ivan

Comment by pin of pic X input ? — May 6, 2007 @ [10:45 pm](#)

42. hi

do you know what R2 connected to VO to VEE

Comment by tom — May 18, 2007 @ [5:31 pm](#)

43. (sorry) on the AGM1264F

Comment by tom — May 18, 2007 @ [5:32 pm](#)

44. #42 & #43 (tom) - I use a 10k potentiometer to adjust the contrast. Otherwise you will have a set contrast. I'm not sure what the optimal voltage divider would be given your situation, but I'd be happy to measure the wiper's position on my board if you would like.

Comment by [Steve](#) — May 18, 2007 @ [6:47 pm](#)

45. thank you so much:)

Comment by tom — May 19, 2007 @ [7:49 pm](#)

46. hi im using solomons graphic display 160*128 dots it has toshiba t693c processor and i ha graphic display yet ... i only used 2line alphanumeric ... so will u please help in providing the pro and displaying data in it .

Comment by ayaz — May 21, 2007 @ [5:37 am](#)

47. Hi
Nice project. I need a little help becouse i'm new to pics. With my display LGM12864B-NSW-BI make to work. I don't understand the glcd.c and where to find it and how to implant in pic18F4550 hex wil be realy good.
Thanks. Zlatko

Comment by Zlatko — June 4, 2007 @ [4:04 pm](#)

48. I maneged to make it wotk but still have some strange problem, the menu (AV, MV) is in the r

Comment by Zlatko — June 4, 2007 @ [5:06 pm](#)

49. You are good , you can consult such a conversion efficiency that CPU is SX28AC , AD is ADC08 Mhz , suppose AD substitute be ADC08231, bandwidth ought to can achieve 10Mhz .sx28's instru PIC16C5X's compatible
SX code all the quickly job, taking advantage of the 50 MIPS.
ADC08831 Resolution: 8 bits ,Conversion time (fC = 2 MHz): 4µs (max),
ADC08231 Resolution: 8 Bits ,Conversion time (fC e 4 MHz) 2 ms (Max),
<http://www.efplus.com/techref/scenix/sxoscope/index.htm>

Comment by han jian (china) — July 21, 2007 @ [8:59 am](#)

50. ADC08231 Resolution: 8 Bits ,Conversion time (fC = 4 MHz) 2 us (Max),

Comment by han jian (china) — July 21, 2007 @ [9:01 am](#)

51. My English is not very good, hoped your PIC18F2550 LCD projects speed may bequicker, congra

Comment by han jian (china) — July 21, 2007 @ [9:05 am](#)

52. can the sampling rate be increased for a higher resolution lcd together with the multiplier (320 x 2

Comment by wildlaurel — August 4, 2007 @ [1:48 pm](#)

53. Steve, i'm use pic18f252 & g64128 from cct and have only a left side of the screen, help me!!

P.S. With simulation on ISIS i'got exectly some picture, with another LCD module - LGM12641E side ...

What can i do?

Sorry for my bad engl.

Comment by Alex Krumholz — August 6, 2007 @ [3:39 pm](#)

54. #46 (ayaz) - Sorry, I am not familiar with the Toshiba T693C. I assume it has a different initialization probably have a different pinout. I unfortunately cannot help you create test code for it because I don't have that utilizes that chipset. Therefore, I would recommend you look for a T693C datasheet and work with it to see what you can do. Good luck!

#47 (Zlatko) - glcd.c is a driver file for CCS PCWH. The reason I cannot post the original GLCD.c is copyrighted to CCS Inc. That means that they own the intellectual property rights to it. In other words, you cannot use the code without their permission. However, if you have CCS PCWH, then you will already have the folder.

#48 (Zlatko) - Great to hear that you got it to work! The reason for the text-in-the-middle is due to have accidentally switched the chip select (CS) lines. Flip them and see if that corrects the display.

#49-51 (han jian (china)) - I am always looking for newer and more efficient ways to capture the data. Your suggestions!

#52 (wildlaurel) - A higher resolution display would allow for more information to be displayed (i.e. more captures); however, the native sampling rate would not increase. A higher speed ADC is necessary for more captures.

#53 (Alex Krumholz) - I have not tested the PIC18F252; however, there is the distinct possibility it is connected to an open-collector output or to an input (which would cause the symptoms you described). I will write some code and see if I can spot anything amiss.

Comment by [Steve](#) — August 7, 2007 @ [9:49 am](#)

55. #53 (Alex Krumholz) - Yep, the CS2 line is connected to RA4, which is an open drain output (see PIC18FXX2 datasheet). This is exactly what I expected (although I did not anticipate it when I posted the code).

One way to correct the problem is to add a pullup resistor between 5V and the output (see http://www.acroname.com/robotics/info/concepts/opn_drct.html). Please let me know if this corrects the problem.

Comment by [Steve](#) — August 7, 2007 @ [10:06 am](#)

56. hi
I need any circuit for pc oscilloscope with USB port.
do you help me?

Comment by rose — August 18, 2007 @ [5:02 pm](#)

57. hello steve very impressive work...i salute you..
i can not understand what is the use of
1. threshold function
2. which compiler i should use.i have programmer no bootloader is required.

Comment by rishi — September 13, 2007 @ [10:30 am](#)

58. Hello steve i build this scope with pic18f4550 and work fine.
In the future i want the build scope with external adc (adc0820, adc08062...) for better performan
Regard.

Comment by Miodrag — September 16, 2007 @ [3:34 pm](#)

59. Hi Steve, I finally got my version of this project working (thanks for your input). Made a simple s
KS0107 during the development stage to allow it to be used with 8 pin PIC's. It's on my projects p
drive it if you or anyone else are interested.

Cheers

Comment by [Gordon](#) — September 16, 2007 @ [6:10 pm](#)

60. Hi Steve,

Just a friendly suggestion - you could use 2 of the ADC channels to make an X/Y mode version of
play back the sound file in the comments to the right of this youtube video to get some nice effects

<http://www.youtube.com/watch?v=s1eNjUgaB-g>

Cheers!

Comment by Dan — September 24, 2007 @ [5:27 am](#)

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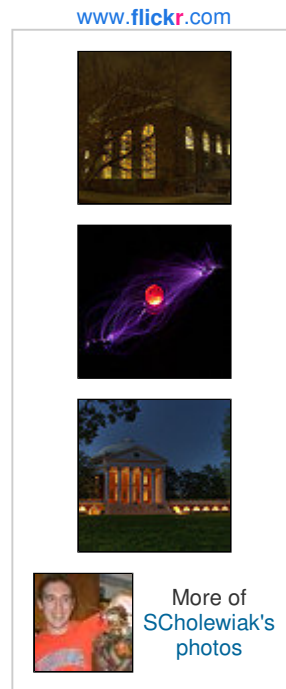
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